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Network-Validated Distributed ADMM Coordination of Cold-Climate Electric Heating under Communication Failures

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ABSTRACT Decentralized coordination of flexible loads through the alternating direction method of multipliers (ADMM) is an established way to flatten the winter peaks that cold-climate electric heating imposes on distribution feeders. Three gaps limit its practical credibility. First, multi-agent schemes are fragile to communication failures: when agents stop responding, the coordinated aggregate degrades in ways rarely quantified. Second, the coordinated aggregate is almost never checked against the physics of the feeder it is supposed to relieve. Third, "comfort" is usually a proxy on shifted energy rather than a check on modelled indoor temperature. We close all three with a network-validated pipeline that combines a comfort-aware sharing-ADMM coordinator, a machine-learning imputer that reconstructs the heating of non-responsive agents, and an AC power-flow validation stage on a synthetic North-American low-voltage residential feeder behind an explicit distribution transformer. On a synthetic Trois-Rivières cold day, the coordination flattens the aggregate peak while holding the worst modelled indoor-temperature excursion to about $1\,^{\circ}$C, roughly half that of a naive energy-only flatten. Crucially, the flattened aggregate clears a transformer thermal overload and restores feeder feasibility rather than merely lowering a statistical peak, and the cleared state stays feasible under heavy communication failure before re-violating at a sharp knee. Comparing three families of machine-learning regressors, we find that whether one imputes at all matters far more than which method. The result is a fully reproducible study that ties coordination quality, thermal comfort, and measurable feeder feasibility together.

INDEX TERMS ADMM, distributed optimization, demand response, thermal comfort, communication failures, machine learning imputation, power flow, low-voltage distribution feeder, distribution transformer, electric heating.

NOMENCLATURE

i	Home (agent) index, $i = 1, \dots, N$.
t	Time-step index, $t = 1, \dots, T$.
N	Number of heating agents (homes).
T	Number of time steps over the day.
$h_{i,t}$	Baseline (thermostat-driven) heating power of home i at step t.
$x_{i,t}$	Coordinated heating power of home i at step t.
$\hat{x}_{i,t}$	Imputed heating power of a non-responsive home i.
B_t	Fixed (non-deferrable) background aggregate load at step t.
c	Constant flat-aggregate target level (energy-conserving).
$\mathbf{1}$	All-ones vector.
α	Per-step comfort deferrability band.
λ	Energy-deviation weight.
γ	Indoor-temperature comfort weight.
μ	Aggregate-flattening weight.
ρ_{ADMM}	ADMM penalty parameter.
ϱ	Non-responsive (communication-failure) fraction of agents.
G_i	First-order RC step-response mapping a heating deviation to home i's indoor-temperature deviation.
z	ADMM auxiliary shared-aggregate variable.
u	ADMM scaled dual variable.
a_i	ADMM penalty anchor for home i, $a_i = z - u$.
M_i	Per-home proximal system matrix in the x-update.
ν_i	Scalar dual of the daily-energy hyperplane in the projection.

$V_{\min}$ Worst-of-day minimum bus voltage (p.u.).

I. INTRODUCTION

THE electrification of heating in cold-climate regions concentrates distribution-system stress into a few winter hours [1]. In the province of Québec, where electric space heating is the dominant heating technology, the system peak is driven almost entirely by simultaneous resistive and heat-pump heating on the coldest mornings and evenings, and field trials confirm that aggregated electric heating sharply raises coincident peak demand [2]. As distributed energy resources (DERs) proliferate and heating loads grow, these synchronized peaks threaten thermal limits on feeder trunk conductors and depress voltages far from the substation, eroding the hosting capacity of low-voltage networks [3], [4]. Demand response that shifts flexible loads to relieve these peaks is the established mitigation [5]. Because the flexibility lives behind thousands of meters and not at a single controllable node, centralized dispatch scales poorly; decentralized coordination, in which each home solves a small local problem and exchanges only an aggregate signal, is the natural fit.

The alternating direction method of multipliers (ADMM) [6] is a workhorse for this setting: its sharing form lets a coordinator flatten a common resource (here, total feeder load) while each agent retains private control of its own heating schedule. ADMM-based demand response is well studied [7], [8], yet two gaps undermine its real-world credibility.

The first gap is fragility to communication failures. Multi-agent coordination implicitly assumes every agent reports on every iteration; when a fraction of agents fall silent, the aggregate the coordinator optimizes is no longer the aggregate that materializes on the wire. How badly the coordinated outcome degrades, and how an operator might recover it by forecasting the missing contributions, is rarely quantified end to end [9], [10].

The second gap is physical validity. The literature typically stops at the coordinated aggregate kilowatt signal and reports peak or peak-to-average reductions. It almost never asks the operational question that matters: is the coordinated aggregate actually feasible on the feeder? A flatter aggregate is worthless if it still overloads a trunk line or sags a remote bus below limits.

This paper closes the loop between coordination and feeder physics. We adapt the sharing-ADMM-plus-imputation methodology from a prior multi-agent thesis and extend it with an AC power-flow validation stage, turning a purely signal-level study into a network-validated one. Our contributions are:

- A comfort-aware sharing-ADMM coordinator for deferrable cold-climate electric heating that flattens the feeder aggregate subject to a per-home indoor-temperature comfort penalty and a daily-energy conservation constraint, so load is shifted rather than curtailed.
- A machine-learning imputation stage that reconstructs non-responsive agents so coordination degrades gracefully as the communication-failure fraction grows; comparing three ML families (regularized-linear, tree-ensemble, and instance-based) against naive and no-imputation baselines, we find that whether one imputes matters far more than which method.
- An AC power-flow validation loop that closes the coordinated aggregate back onto feeder physics, quantifying the restored transformer and voltage feasibility and its breakdown under a swept non-responsive fraction with Monte-Carlo risk bands.

The remainder of the paper is organized as follows. Section IV is preceded by the related work and the system model; Section II surveys ADMM-based demand response, multi-agent resilience, telemetry imputation, and distribution power flow, and states our delta against each. Section III formalizes the agents, the feeder, and the communication-failure model. Section IV develops the sharing-ADMM updates, the ML imputer, and the power-flow validation loop. Section V details the case-study configuration, Section VI reports the aggregate, robustness, and network-feasibility results, and Section VII concludes.

II. RELATED WORK

Our contribution sits at the intersection of five strands of literature: distributed (ADMM/consensus) coordination, network-aware demand response, resilience of distributed coordination to communication failures, machine-learning imputation of missing load telemetry, and the flexibility of residential electric heating. We review each and then make the gap explicit (Table 1).

A. ADMM AND CONSENSUS FOR DISTRIBUTED ENERGY COORDINATION

ADMM [6] decomposes a coupled optimization into local subproblems linked by a consensus or sharing constraint, and is now a standard tool for distributed energy coordination. The sharing form (Boyd et al., Section 7.3) is particularly suited to demand flattening, where many agents contribute to one shared resource. It underpins distributed optimal power flow [11], communication-efficient and consensus-based demand response [7], [8], asynchronous microgrid energy management [12], and hierarchical transactive coordination of home energy management systems [13]. These works establish that ADMM converges to a flattened or cost-optimal aggregate; with few exceptions they treat the aggregate signal as the deliverable and do not verify that it is feasible on the underlying feeder.

B. NETWORK-AWARE COORDINATION

A second strand embeds network physics in the coordination itself. Distributed AC optimal power flow couples ADMM with the branch-flow equations [14], and recent work coordinates distributed energy resources with inverter Volt-VAr control through a three-block AC-OPF ADMM

[15]. Network-constrained demand-response pricing [16], ADMM-based congestion-management markets [17], and capacity-constrained electric-vehicle charging [18] all respect feeder limits during coordination. A parallel line handles uncertainty through chance-constrained AC-OPF, enforcing probabilistic voltage and thermal limits for renewables [19] and flexible-load ensembles [20]. This literature proves that network-feasible coordination is achievable, but each work assumes perfect, complete communication: every agent reports on every iteration.

C. RESILIENCE TO COMMUNICATION FAILURES

A third strand studies how coordination degrades when agents fall silent. Brown et al. show that even a single lost link can drive multi-agent systems to arbitrarily poor states [9], and fault-tolerant consensus is a mature survey topic [21]. In distributed optimization, stochastic communication delay slows ADMM-based OPF [10], lossy links are handled by robust dispatch [22] and relaxed asynchronous ADMM with convergence guarantees [23], and adversarial or non-responsive agents can be detected and isolated to preserve convergence [24]. Learning has begun to enter the loop, e.g. to warm-start consensus-ADMM OPF [25]. This literature characterizes when coordination breaks and offers robust-by-design or detect-and-isolate responses, but it rarely reconstructs a silent agent's contribution, and never propagates that reconstruction error into feeder physics.

D. IMPUTATION AND FORECASTING OF MISSING TELEMETRY

When an agent falls silent the coordinator must estimate its consumption. Machine-learning imputation of missing smart-meter data spans denoising diffusion models [26], correlation-clustering for faulted grid data [27], and low-rank recovery of asynchronous meter streams [28]. For the closely related task of short-term load prediction, gradient-boosted trees are competition-winning regressors [29], [30], one of the families we evaluate [31] for imputing a silent home's heating from weather and calendar features. Crucially, in all of this work the imputer or forecaster is evaluated in isolation, by held-out RMSE, and is never embedded in a distributed coordinator nor propagated into a power-flow feasibility check.

E. RESIDENTIAL ELECTRIC-HEATING FLEXIBILITY

Finally, the resource we coordinate, thermostatically controlled electric heating, is itself well studied. Aggregate thermostatic flexibility can be quantified for demand response [32], direct electric space heating combined with partial thermal storage shifts load under comfort constraints via linear programming [33], heat-pump-plus-storage flexibility reduces peak demand and generation capacity [34], and field experiments confirm the aggregate effect and its rebound dynamics [35]. In cold climates specifically, electrically heated buildings with thermal storage enable peak shaving [36], and multi-unit residential flexibility and predictive heating control have been demonstrated in Québec-style winter-peaking contexts [37], [38]. The distributed-control variants closest to ours coordinate large TCL populations through asynchronous greedy [39] or resilient-consensus [40] schemes; yet these remain abstract control formulations without an AC feeder model, and the broader thermal-DR literature is overwhelmingly centralized and not network-validated.

F. THE GAP

Table 1 positions representative prior work along the five capabilities our study requires together: distributed ADMM/consensus coordination, AC power-flow validation, an explicit communication-failure model, machine-learning imputation of the missing agents, and uncertainty quantification of the resulting risk. Every ingredient exists, but only in disjoint subsets. Distributed-ADMM coordinators assume perfect communication; network-aware and chance-constrained methods are centralized or assume complete reporting; resilience methods detect or tolerate failures rather than reconstruct the missing agents; imputation accuracy is studied standalone; and thermal-DR studies seldom touch the feeder at all. To our knowledge no prior work closes the full loop: distributed ADMM coordination of cold-climate electric heating, machine-learning imputation of communication-failed homes, AC power-flow validation on a distribution feeder, and Monte-Carlo quantification of the resulting violation probability under a swept non-responsive fraction. That closed loop is the contribution of this paper.

III. SYSTEM MODEL

This section defines the model in general terms; the population size, horizon, weights, feeder ratings, tariff, and sweep used to instantiate it are collected in Section V. Figure 1 gives the closed-loop pipeline that the rest of the paper develops.

A. AGENTS AND DEFERRABLE HEATING

We coordinate a population of N representative cold-climate electric-heating homes over a single cold day discretized into T sub-hourly steps. Each home i has a thermostat-driven baseline heating profile $h_{i,t}$ (produced by a behavioral model conditioned on the ambient temperature) plus a fixed, non-deferrable background load. Heating is deferrable within a comfort band of $\pm\alpha$ around the baseline: the coordinator may shift when a home heats but conserves its total daily heating energy, so demand is shifted, not curtailed. With $x_{i,t}$ the coordinated heating power of home i at step t, each agent is constrained by

$$(1-\alpha)\, h_{i,t} \leq x_{i,t} \leq (1+\alpha)\, h_{i,t}, \qquad \sum_t x_{i,t} = \sum_t h_{i,t}. \tag{1}$$

The box bound limits the per-step re-timing and the equality plane conserves each home's daily heating energy. Thermal comfort, however, is not a per-step energy quantity

TABLE 1. Positioning against representative prior work. ✓: addressed; ∼: partial; –: not addressed. PF: power flow.

Representative work	Distributed ADMM/consensus	AC PF validation	Comms-failure model	ML imputation of missing agents	Uncertainty quantification
Distributed AC-OPF [11], [14]	✓	✓	–	–	–
ADMM AC-OPF DER coord. [15]	✓	✓	–	–	–
Consensus / sharing DR [8], [13]	✓	–	–	–	–
Chance-constrained OPF [19], [20]	–	✓	–	–	✓
ADMM-OPF under delay [10]	✓	✓	✓	–	∼
Resilient distributed opt. [24]	✓	–	✓	–	–
Learning-augmented ADMM-OPF [25]	✓	✓	–	∼	–
Distributed TCL control [39], [40]	✓	–	∼	–	–
ML telemetry imputation [26], [28]	–	–	–	✓	–
Thermal-DR flexibility [32], [33]	–	–	–	–	–
This work	✓	✓	✓	✓	✓

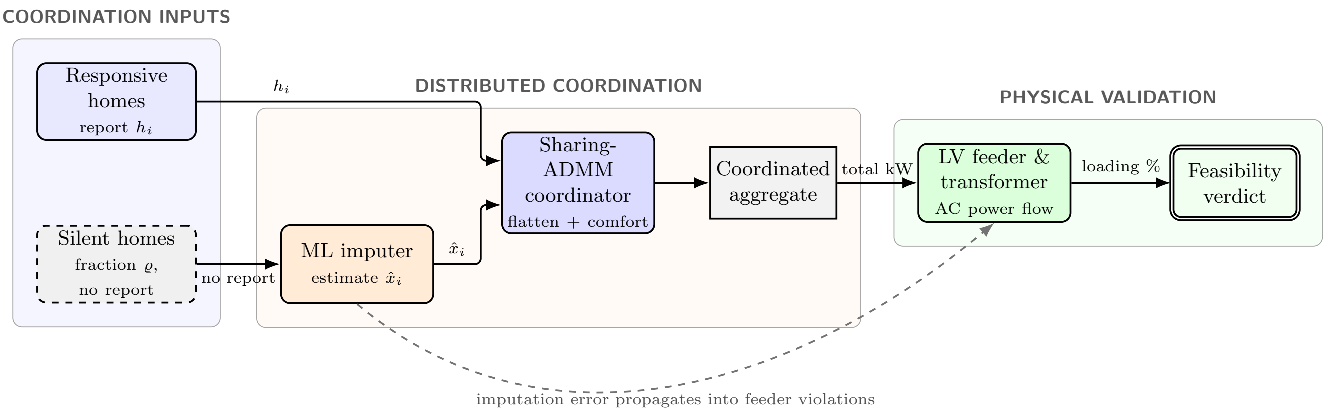

FIGURE 1. Closed-loop pipeline. Responsive homes report their heating; a fraction ϱ fall silent and are reconstructed by a machine-learning imputer. The sharing-ADMM coordinator flattens the aggregate subject to comfort, and the coordinated aggregate is injected into the LV feeder and validated by AC power flow against the transformer thermal limit. The dashed path is the contribution this study quantifies: how imputation error propagates into feeder violations.

but the resulting indoor-temperature excursion: re-timing heat changes when a home is warm, and only a thermal model exposes that. We therefore model comfort physically. For home i a first-order RC step-response G_i maps a heating deviation $x_i - h_i$ to the home's indoor-temperature deviation from setpoint, $G_i(x_i - h_i)$; a high-thermal-mass home has a more damped G_i and can absorb re-timing with little temperature swing. This modelled excursion, not the shifted energy, is the comfort quantity the coordinator controls (Section IV) and that we report. Background load is held fixed and is not subject to (1).

B. THE FEEDER

Physical feasibility is assessed on a synthetic low-voltage (LV) residential feeder with an explicit MV/LV distribution transformer, modeled in pandapower [41]. The homes are connected as literal loads (no aggregation and no artificial scaling) behind the transformer and distributed across several parallel LV feeders, so their aggregate heating loads the transformer directly. The binding constraint is therefore the transformer's thermal loading, the classic residential winter-peak (cold-load-pickup) limit, assessed against a short-duration overload limit; LV bus voltage is reported as a secondary severity measure. All loads use a common lagging power factor. The one-line topology is shown in Figure 2.

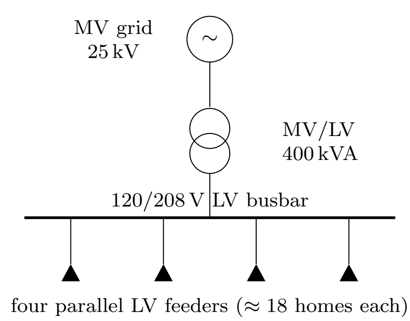

FIGURE 2. One-line diagram of the synthetic LV residential test feeder: an MV grid source, a single MV/LV distribution transformer, and an LV busbar feeding four parallel radial feeders that carry the homes as literal loads.

C. COMMUNICATION-FAILURE MODEL

At coordination time a fraction ϱ of the agents become non-responsive and report nothing to the coordinator; we sweep ϱ across the unit interval. We use ϱ for this communication-failure fraction throughout, kept notationally distinct from the ADMM penalty ρ_{ADMM} of Section IV. For each non-responsive agent the coordinator substitutes an imputed heating profile $\hat{x}_{i,t}$ produced by a trained regressor (Section IV) and proceeds as if it were the true contribution. The imputation error therefore propagates first into the coordinated aggregate and ultimately into the feeder power flow: exactly the propagation path this study quantifies.

IV. METHODOLOGY

This section states the method in general terms; the concrete feeder, agent population, weights, and tariff used to instantiate it are given in Section V.

A. SHARING-ADMM COORDINATION

Let $x_i \in \mathbb{R}^T$ be the heating schedule of home i, h_i its baseline, and $B \in \mathbb{R}^T$ the fixed background aggregate. The coordinator flattens the total feeder load toward a constant target c while respecting each home's comfort. We solve the sharing problem

$$\begin{aligned} \min_{\{x_i\}} \; & \sum_{i=1}^{N} \Big[\tfrac{\lambda}{2}\|x_i - h_i\|_2^2 + \tfrac{\gamma}{2}\|G_i(x_i - h_i)\|_2^2 \Big] \\ & + \; \tfrac{\mu}{2}\Big\| \sum_{i=1}^{N} x_i + B - c\,\mathbf{1} \Big\|_2^2 \end{aligned} \tag{2}$$

subject to (1) for every i, where c is the constant level that conserves total daily energy, the weight λ penalizes energy deviation, the weight γ penalizes the modelled indoor-temperature excursion, the weight μ penalizes departure of the total load from a flat profile, and $\mathbf{1}$ is the all-ones vector. Here G_i is the home's first-order RC step-response mapping a heating deviation to the home's indoor-temperature deviation, so the per-home penalty $\frac{\gamma}{2}\|G_i(x_i - h_i)\|_2^2$ bounds the cumulative temperature swing rather than only the per-step energy shift; because G_i is more damped for high-thermal-mass homes, this term steers re-timing toward homes that can absorb it. The first two terms keep each home close to its baseline in both energy and comfort; the third rewards a flat aggregate. Because daily energy is conserved, c is a constant and the aggregate term is exactly the variance of the total load; the scheme is valley-filling, not curtailment.

Following the sharing form of ADMM [6], we introduce an auxiliary aggregate variable z and a scaled dual u with penalty parameter ρ_{ADMM}, and iterate three steps. The **x-update** solves each home's local problem in parallel: a comfort-plus-penalty proximal step followed by projection of the result onto the box-and-energy-plane feasible set of (1). The **z-update** is the closed-form average that drives the shared aggregate toward the flat target. The **u-update** is the standard scaled dual ascent on the sharing residual. We tune the penalty ρ_{ADMM} and apply over-relaxation [6] to the primal update, which keeps the comfort-coupled iteration count low. Algorithm 1 summarizes one full sweep.

Algorithm 1: Sharing-ADMM heating coordination

Input: baselines $\{h_i\}$, background B, target c, weights λ, μ, penalty ρ_{ADMM}
Output: coordinated schedules $\{x_i\}$
initialize $x_i \leftarrow h_i$, $z \leftarrow 0$, $u \leftarrow 0$;
repeat
  foreach *home i (in parallel)* **do**
    proximal step on $\frac{\lambda}{2}\|x_i - h_i\|^2$ plus penalty toward $z - u$;
    project x_i onto the box-and-energy set of (1);
  $z \leftarrow$ closed-form average enforcing the flat aggregate target;
  $u \leftarrow u + \big(\sum_i x_i + B\big) - z$;
until *aggregate residual converged*;
return $\{x_i\}$;

The x-update projection is what makes the scheme physically meaningful: the box conserves daily energy, while the indoor-temperature penalty γ bounds the modelled comfort excursion, so the coordinated schedule is feasible for the homes and gentle on their indoor temperature. The number of iterations to convergence is recorded as a tractability metric; the temperature-coupling term slows consensus, an effect we quantify in Section VI.

1) Capped-energy projection in the x-update

We make the local step explicit because its structure is what keeps every iterate feasible. For home i the unconstrained comfort-plus-penalty objective is a strictly convex quadratic, so its minimizer is available in closed form. Writing $a_i = z - u$ for the current penalty anchor, the prox optimum solves the linear system

$$\begin{aligned} \tilde{x}_i &= M_i^{-1}\big[(\lambda I + \gamma\, G_i^\top G_i)\, h_i + \rho_{\mathrm{ADMM}}\, a_i\big], \\ M_i &= (\lambda + \rho_{\mathrm{ADMM}})\, I + \gamma\, G_i^\top G_i, \end{aligned} \tag{3}$$

which trades the energy-plus-comfort pull toward h_i against the consensus pull toward a_i. Because G_i is fixed per home, the inverse M_i^{-1} is precomputed once, so the temperature-aware prox stays closed-form and cheap per sweep; with $\gamma = 0$ it collapses to the per-step convex combination of the energy-only scheme. This unconstrained point need not respect (1), so we Euclidean-project it onto the intersection of the comfort box $[(1-\alpha)h_{i,t}, (1+\alpha)h_{i,t}]$ and the daily-energy hyperplane $\sum_t x_{i,t} = \sum_t h_{i,t}$:

$$x_i \;=\; \arg\min_{x} \; \tfrac{1}{2}\|x - \tilde{x}_i\|_2^2 \quad \text{s.t.} \quad x \in \mathrm{box}_i, \;\; \mathbf{1}^\top x = \mathbf{1}^\top h_i. \tag{4}$$

Introducing a single scalar dual ν_i for the energy constraint, the projection has the water-filling form $x_{i,t} = \mathrm{clip}_{[(1-\alpha)h_{i,t},\,(1+\alpha)h_{i,t}]}(\tilde{x}_{i,t} + \nu_i)$, and the value of ν_i that restores $\mathbf{1}^\top x_i = \mathbf{1}^\top h_i$ is found by a 1-D bisection on the scalar dual. Because the clipped sum is monotone non-decreasing in ν_i, the bisection is exact and costs only a handful of evaluations, so the whole x-update is closed-form up to a scalar root-find. The subsequent z-update is fully separable per time step (each component of z is the average of the corresponding aggregate-plus-dual component driven toward the flat target c) and therefore also closed form. Both updates are vectorized across agents, so a full sweep remains inexpensive as the home population grows.

B. IMPUTATION OF NON-RESPONSIVE AGENTS

When agent i is non-responsive its baseline h_i is unavailable to the coordinator, so we replace it with a learned estimate $\hat{h}_i$. We train a regressor that maps a small feature vector (ambient temperature, the sine and cosine of the hour of day, and the agent's nameplate heating level) to instantaneous heating power. The model is trained on the responsive population and applied to each silent agent, yielding $\hat{x}_{i,t}$ that the coordinator treats as the agent's contribution. We keep the estimator generic and compare three families of machine-learning regressors, namely a regularized linear model (ridge regression), tree ensembles (gradient boosting [31] and random forests), and an instance-based model (k-nearest neighbours), against naive and no-imputation baselines in Section VI, and report the most accurate. Generalization to unseen homes is assessed by leave-agents-out cross-validation, which holds out entire agents rather than random samples and therefore measures the realistic case of imputing a home the model never saw.

C. POWER-FLOW VALIDATION LOOP

The novel stage closes the loop on physics. For each scenario (uncoordinated, ideal, and each non-responsive fraction ϱ) we inject the resulting aggregate into a radial distribution-feeder model with an explicit distribution transformer, distributing the demand across the feeder's load points, and solve an AC power flow at every time step. Per scenario we record the worst-of-day maximum transformer loading and minimum bus voltage. A scenario is flagged as violating if the transformer exceeds its thermal rating at any step. This turns an abstract kilowatt signal into a binary feeder verdict and a pair of continuous severity measures.

Two modeling choices are methodological rather than case-specific. Sizing the binding constraint into overload is what makes the experiment decision-relevant: a feeder comfortably within limits in the uncoordinated case has nothing to gain from demand response, so the transformer is rated such that the uncoordinated winter peak is in overload, placing the binding constraint exactly where coordination must act to clear it. Solving every step rather than only the aggregate-peak step matters because imputation error can shift the worst instant: the uncoordinated worst is the evening peak, but under heavy imputation the binding step can migrate, and an aggregate-peak-only check would miss it. Reporting the worst-of-day envelope across the whole horizon therefore gives a conservative, instant-agnostic verdict.

D. SOLVER VALIDATION

Because the x-update, z-update, and projection are hand-rolled, we validate the distributed ADMM against an independent centralized solve of the same convex quadratic program with a general-purpose convex solver (CVXPY). The optimality gap of the distributed solver is reported in Section VI.

E. UNCERTAINTY QUANTIFICATION

The deterministic sweep fixes one failing-agent subset and the raw point forecast per fraction. To quantify risk we run a Monte-Carlo analysis over two sources of uncertainty: which agents fall silent (drawn at random) and the imputation error (the silent agents' estimates are perturbed by Gaussian residuals matched to the imputer's cross-validated error). For each fraction we draw many such realizations, re-solve the coordination, and record the distribution of the daily peak. Because the worst-of-day network metrics are monotone functions of that peak (the transformer sees the whole aggregate regardless of spatial placement), each realization's peak is mapped through a precomputed peak-to-loading curve, making the Monte-Carlo sweep inexpensive while reproducing the deterministic operating points exactly. We report the mean, a central band, and the probability that the transformer is in violation.

V. CASE STUDY SETUP

The study runs on a single synthetic Trois-Rivières cold day with a minimum ambient temperature of $-21.9\,°\mathrm{C}$, discretized at 15-minute resolution into $T = 96$ steps. We coordinate $N = 74$ electric-heating homes with comfort band $\alpha = 0.5$, energy weight $\lambda = 0.15$, indoor-temperature comfort weight $\gamma = 2$, flattening weight $\mu = 1.0$, and ADMM penalty $\rho_{\mathrm{ADMM}} = 5$ with over-relaxation factor 1.7 (tuned so the comfort-coupled solver converges in well under one hundred sweeps). The communication-failure fraction is swept from $\varrho = 0$ to $\varrho = 1.0$ over twelve points, $\varrho \in \{0, 0.2, 0.4, 0.6, 0.7, 0.8, 0.82, 0.85, 0.88, 0.9, 0.95, 1.0\}$, sampled densely near the feasibility knee at $\varrho \approx 0.85$.

Energy cost is evaluated against a time-of-use tariff with an off-peak rate of 0.078, a mid-peak rate of 0.128, and an evening-peak rate of 0.158 CAD/kWh. Physical feasibility uses a synthetic North-American LV residential feeder of 74 literal all-electric dwellings behind a single 400 kVA MV/LV network transformer, stepping the Hydro-Québec 25 kV (14.4/24.94 kV) primary down to the North-American 120/208 V three-phase secondary (0.208 kV), split across four parallel LV feeders, with all loads at a power factor of 0.95 and modeled one-for-one (no artificial scaling). This single-transformer service represents a dense multi-unit residential cluster rather than detached homes on individual split-phase transformers. The uncoordinated

TABLE 2. Principal case-study parameters.

Parameter	Value
Homes N	74
Horizon / resolution	96 steps / 15 min
Coldest-day min. temperature	$-21.9\,°\mathrm{C}$
Comfort band α	0.5
Energy weight λ	0.15
Comfort weight γ	2
Flattening weight μ	1.0
ADMM penalty ρ_{ADMM}	5
ADMM over-relaxation	1.7
Non-responsive fraction ϱ	0 to 1.0, 12 points (dense near 0.85)
TOU off / mid / peak (CAD/kWh)	0.078 / 0.128 / 0.158
Feeder	Synthetic LV residential, 120/208 V
Transformer	400 kVA MV/LV network (25 kV/0.208 kV)
LV feeders	4 parallel
Transformer limit	105% (IEEE C57.91 short-duration)
Power factor	0.95
Uncoordinated transformer loading	109.6%
Power flow	AC Newton-Raphson
Software	pandapower 3.4.0, scikit-learn 1.6.1, LightGBM 4.6.0

coincident peak loads the transformer to 109.6%. A scenario is flagged as violating when the transformer exceeds a 105% loading limit rather than its 100% nameplate: IEEE C57.91 permits short-duration overload because cold ambient air and the transformer's thermal mass let it carry a winter peak above nameplate for a limited time, so 105% is a conservative short-duration winter-loading limit rather than the unrealistically strict nameplate cliff. The software stack is pandapower 3.4.0 for power flow, scikit-learn 1.6.1 for the cross-validation harness, and LightGBM 4.6.0 for the imputer. Table 2 collects the principal parameters, and the reproducibility of the run is discussed below.

A. REPRODUCIBILITY

The study is designed to be reproduced exactly. It depends on no proprietary or field-measured data: the cold day, per-home thermal baselines, and background loads are all synthetic and generated from fixed random seeds, so the inputs are defined entirely by the parameters in Table 2 and regenerate identically on any machine. The computation is a single automated pipeline (Table 3) that runs end-to-end from that configuration without manual intervention, and it is built on open-source numerical libraries (pandapower for power flow, scikit-learn and LightGBM for the imputer), so no closed component stands between the configuration and the reported numbers. Each stage writes its outputs with an integrity digest, every figure and table in this paper is generated from those recorded outputs rather than transcribed by hand, and a regression check pins the headline quantities so that an accidental change is caught. The implementation and configuration are released with the paper. We claim re-runnable reproducibility on this configuration, not statistical generality across feeders or climates, which the limitations discuss.

TABLE 3. Computational pipeline (principal stages).

#	Stage	Principal output
1	Weather generation	Synthetic cold-day temperature
2	Baseline heating	Per-home $h_{i,t}$ profiles
3	Background load	Fixed non-deferrable aggregate B
4	ADMM coordination	Ideal $x_{i,t}$, iteration count
5	Imputer training	Estimator + cross-validation metrics
6	Communication sweep	Per-ϱ imputed aggregates
7	Power-flow validation	$V_{\min}$, transformer loading, verdict
8	Uncertainty + comparison	Violation probabilities, method ranking

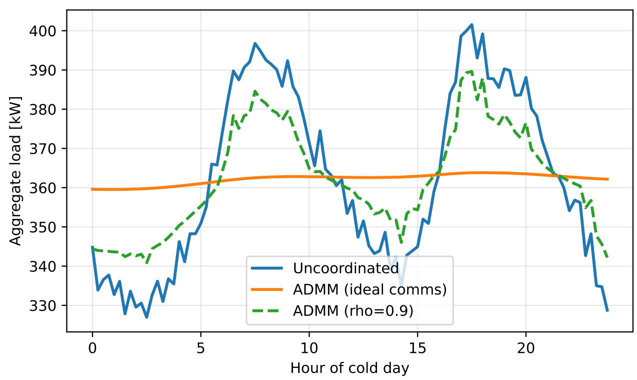

FIGURE 3. Aggregate cold-day load: uncoordinated double-peak versus the flat ADMM-ideal profile (≈ 363.8 kW) and the $\varrho = 0.8$ imputed case with residual peaks. Here ϱ is the non-responsive fraction.

VI. RESULTS

A. AGGREGATE COORDINATION

Figure 3 shows the cold-day aggregate load. The uncoordinated profile exhibits the classic double peak (a morning and an evening heating surge), whereas the ideal ADMM coordination flattens it to a nearly constant ≈ 363.8 kW. The $\varrho = 0.8$ trace, in which most agents are imputed, recovers much of the flattening but leaves residual peaks where imputation error accumulates.

Table 4 quantifies the aggregate key performance indicators. Ideal coordination cuts the peak from 401.6 to 363.8 kW, a 9.4% reduction, and drives the peak-to-average ratio (PAR) from 1.109 down to 1.004, essentially a flat load. A PAR of 1.004 is near the practical floor: the coordinated aggregate is constant to within half a percent across the entire day, so the flattening objective in (2) has been met almost exactly even with the comfort term steering re-timing toward high-thermal-mass homes. Energy cost falls slightly, from 909.5 to 896.1 CAD (a 1.48% reduction), because the daily-energy constraint conserves total heating; the saving comes from shifting load out of the evening-peak TOU window (0.158 CAD/kWh) toward off-peak hours (0.078 CAD/kWh) rather than from using less energy. The modest size of the cost reduction is itself a useful result, and an honest one: the objective in (2) flattens the aggregate (it penalizes departure from a constant level), it does not chase the TOU price signal. A purely flat profile still places substantial load in the priced evening window, so the bill moves only as a side effect of peak-shaving. With energy conserved, demand response of this kind buys capacity relief (a lower peak) far more than it buys energy savings, and the case for it must rest on the network benefit rather than the bill.

B. THERMAL COMFORT

The flattening is achieved without sacrificing thermal comfort, but establishing that requires the physical comfort model, not an energy proxy. We validate it by propagating each home's coordinated schedule through its RC thermal model and recording the worst modelled indoor-temperature excursion from setpoint (Figure 4). The comfort-aware coordination ($\gamma = 2$) holds the worst excursion across all 74 homes to $1.01\,°\mathrm{C}$, with a median of only $0.70\,°\mathrm{C}$, inside a $1\,°\mathrm{C}$ comfort band and comparable to a thermostat's own deadband. An energy-only flatten that ignores the temperature term, by contrast, pushes the worst excursion to $2.11\,°\mathrm{C}$: a clearly noticeable swing.

This is exactly the gap the per-step box cannot close. Keeping every step inside the $\pm 50\%$ band and conserving daily energy bounds the instantaneous power shift but not the cumulative temperature excursion, which integrates the heating deficit through the home's thermal time constant; only the RC-coupled penalty $\frac{\gamma}{2}\|G_i(x_i - h_i)\|_2^2$ bounds that integral and steers re-timing toward the homes that can absorb it. The earlier claim that comfort was "preserved by construction" by the box alone was therefore optimistic. Crucially, the comfort term costs almost no flattening: the comfort-aware peak reduction is 9.4% against 9.8% for the energy-only flatten, so halving the indoor-temperature excursion sacrifices only about half a percentage point of peak shaving.

1) A plateau-and-ramp transition

Swept finely (Table 4), the degradation of coordination with the non-responsive fraction is not a smooth decay but a plateau followed by a ramp. From $\varrho = 0$ to $\varrho \approx 0.7$ the coordinated peak stays near the ideal optimum (363.8–367.2 kW); beyond a knee near $\varrho = 0.87$ it rises toward the uncoordinated peak, reaching 375.0, 383.1, 385.9, 389.6, and finally 401.9 kW at $\varrho = 0.8, 0.85, 0.88, 0.9,$ and 1.0. This shape is the signature of a flexibility-budget mechanism, and it has two causes. First, an imputed home is pinned to its forecast, which reproduces its natural, peaky heating shape, not a flattened one, so it leaves the coordination pool; degradation is therefore driven mainly by the loss of controllable flexibility, not by raw forecast error: the per-scenario imputation RMSE stays small (0.23–0.37 kW, last column) and rises far more slowly than the peak excess. Second, the remaining responsive homes flatten

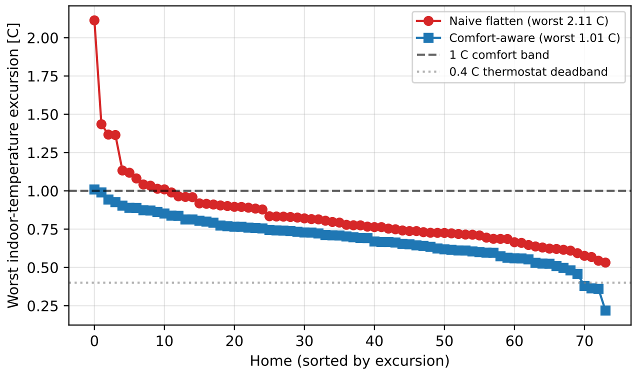

FIGURE 4. Per-home worst modelled indoor-temperature excursion from setpoint, comfort-aware coordination ($\gamma = 2$) versus a naive energy-only flatten, propagated through each home's RC thermal model. The comfort-aware worst is $1.01\,°\mathrm{C}$ (median $0.70\,°\mathrm{C}$) against $2.11\,°\mathrm{C}$ for the energy-only case; the $1\,°\mathrm{C}$ comfort band and the $0.4\,°\mathrm{C}$ thermostat deadband are marked.

TABLE 4. Aggregate KPIs by non-responsive fraction ϱ (selected points; the full 12-point sweep appears in Figure 5). PAR and cost are reported for the two headline scenarios; RMSE is the per-scenario imputation error.

Scenario	Peak (kW)	PAR	Cost (CAD)	RMSE (kW)
Uncoordinated	401.6	1.109	909.5	—
Coordinated ideal	363.8	1.004	896.1	—
ϱ=0.4	364.9	—	—	0.231
ϱ=0.6	365.9	—	—	0.245
ϱ=0.7	367.2	—	—	0.240
ϱ=0.8	375.0	—	—	0.367
ϱ=0.85	383.1	—	—	0.360
ϱ=0.88	385.9	—	—	0.355
ϱ=0.9	389.6	—	—	0.352
ϱ=1.0	401.9	—	—	0.363

the total only up to their collective $\pm 50\%$ deferral budget: as long as that budget exceeds the peakiness injected by the pinned population, the peak is held near the optimum (the plateau); once it is exhausted, around $\varrho \approx 0.87$, the residual peak passes through. The transition is thus sharp at its onset (a binding-constraint knee) but smooth thereafter.

C. IMPUTATION ACCURACY AND ROBUSTNESS

The deployed ML imputer (gradient-boosted trees) is a competent forecaster of an unseen home's heating. Under 5-fold leave-agents-out cross-validation it achieves an RMSE of 0.304 kW and an MAE of 0.255 kW, against a heating standard deviation of 0.736 kW. At roughly 41% of the natural variability, the imputer is clearly better than predicting the mean and is accurate enough that the coordinated aggregate barely moves for non-responsive fractions below the flexibility knee, as seen in the per-scenario RMSE column of Table 4. Notably the per-scenario imputation RMSE stays small across the sweep (0.23–0.37 kW, well under the 0.736 kW heating spread) and rises only modestly even as the aggregate peak degrades sharply past the knee: direct evidence that the damage comes from the loss of controllable agents, not from worsening forecasts.

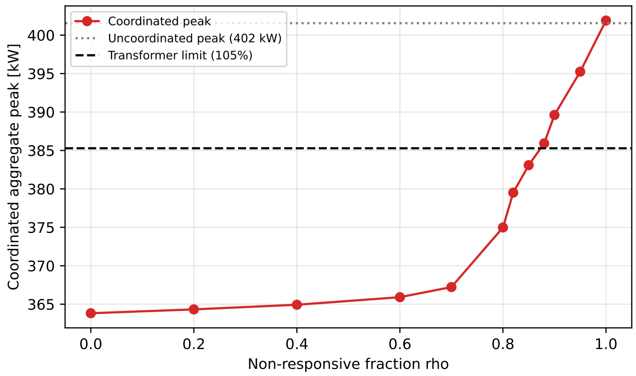

FIGURE 5. Coordinated aggregate peak versus the non-responsive fraction ϱ, against the uncoordinated level and the peak that corresponds to the transformer thermal limit. The horizontal axis labelled "rho" in the plotted image denotes this fraction ϱ, not the ADMM penalty ρ_{ADMM}.

Figure 5 plots the coordinated peak against ϱ over the full 12-point sweep. (We show the peak alone: because daily energy is conserved the mean load is constant, so the peak-to-average ratio is proportional to the peak and carries no additional information.) The peak stays near the optimum through $\varrho \approx 0.8$ and then rises toward the uncoordinated value at $\varrho = 1$: a plateau-and-ramp, not a gradual decay, with the ramp onset (knee near $\varrho \approx 0.87$) marking exhaustion of the responsive deferral budget. The curve crosses the transformer-limit-equivalent peak near $\varrho = 0.88$.

The temperature-coupling term raises the conditioning of each home's x-update, since it couples the time steps of the local prox through $G_i^\top G_i$. Left unaddressed this would slow consensus, so we tune the ADMM penalty ρ_{ADMM} and apply over-relaxation [6] to the primal update. With these, the comfort-coupled solver stays fast: the energy-only scheme converges in ≈ 21 sweeps and the comfort-aware ideal case in ≈ 30, rising monotonically to ≈ 99 at the highest non-responsive fractions but always reaching the residual tolerance well within the iteration cap. The iteration count therefore grows only mildly with the non-responsive fraction; we report it as a tractability diagnostic, not an operational early-warning signal.

D. IMPUTATION METHOD COMPARISON AND THE VALUE OF IMPUTATION

The sweeps so far use one representative ML imputer. Two questions remain: how much does imputation itself buy, and how much does the choice of method matter? To answer them fairly we adopt a stricter realized-load model that separates belief from reality: a silent home physically draws its true (uncontrolled, peaky) heating, while the coordinator sees only an estimate and flattens the responsive homes against it; the feeder then experiences the realized aggregate: responsive coordinated schedules plus the true silent load plus background. "No imputation" is the coordinator flattening the responsive subset blind to the silent homes.

TABLE 5. Imputation methods: intrinsic accuracy (leave-agents-out CV) versus realized feeder impact at $\varrho = 0.8$ (mean over 200 random silent subsets) under the 105% limit. Lower realized loading and violation probability are better.

Method	RMSE (kW)	R^2	Peak (kW)	Load. (%)	P_{viol}
Ridge	0.280	0.70	374.8	102.1	0.00
k-NN	0.308	0.67	374.9	102.1	0.00
LightGBM	0.304	0.67	378.0	102.9	0.10
Random forest	0.342	0.62	377.4	102.8	0.09
Mean-level (naive)	0.312	0.65	374.7	102.0	0.00
No imputation	—	—	395.8	107.9	1.00

We compare three families of learned regressors, namely a regularized linear model (ridge regression), tree ensembles (gradient boosting and random forests), and an instance-based model (k-nearest neighbours), together with a naive "mean-level" baseline (a flat profile at each silent home's known nameplate level, with correct energy but no shape) and the no-imputation case.

Table 5 reports both layers of metric. Intrinsically (5-fold leave-agents-out), ridge regression is the most accurate (RMSE 0.280 kW, $R^2 = 0.70$), slightly ahead of LightGBM (0.304 kW) and k-NN (0.308 kW) and the random forest (0.342 kW); with this small population and four smooth features the linear model is hard to beat. Notably the naive mean-level baseline is close to the learned methods on RMSE (0.312 kW), because each home's heating is dominated by its mean level, which that baseline gets exactly right; so intrinsic RMSE alone badly under-rates the importance of getting the timing right.

The realized-impact columns and Figure 6 tell the operational story, and here the value of imputation is decisive. Without imputation the realized peak climbs to 395.8 kW (107.9% loading) at $\varrho = 0.8$ and the transformer violates the 105% limit in every draw ($P_{\mathrm{viol}} = 1.00$), because the outcome depends entirely on which uncontrolled homes fall silent. Every imputer, by contrast, holds the feeder cleared: the realized peak stays near 375–378 kW ($\approx$ 102–103%) and the violation probability collapses to at most 0.10 (LightGBM) and to 0.00 for ridge, k-NN, and the naive mean-level baseline. The dominant gap is therefore imputation versus none, roughly an 18–21 kW reduction (396 to 375–378 kW) that turns a certain violation into a cleared feeder, and the choice of method is secondary at this fraction. Among the methods the ranking still tracks intrinsic accuracy: the regularized linear model (ridge) is best both intrinsically (RMSE 0.280 kW, $R^2 = 0.70$) and on realized loading (102.1%, $P_{\mathrm{viol}} = 0.00$), ahead of the instance-based and tree-ensemble methods, with the gradient-boosted model the only one carrying non-trivial residual risk (0.10). Ridge wins for a physically intuitive reason. A home's heating is, to first order, a linear function of these features: steady-state demand scales with the indoor–outdoor temperature gap and with the home's nameplate level, with only a mild diurnal term, so the target is smooth and nearly

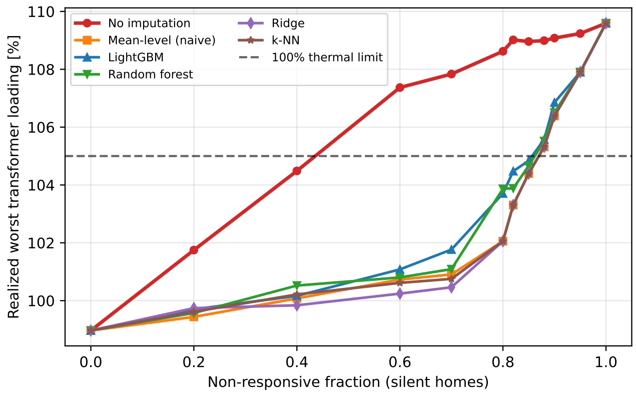

FIGURE 6. Realized worst transformer loading versus the non-responsive fraction for each imputation method under the realized-load model, against the 105% short-duration limit. No imputation (red) diverges early and breaches the limit; the imputers (learned and naive) cluster near 102–103% and stay cleared until the knee. All converge to the uncoordinated 109.6% once every home is silent.

linear. With just a few dozen homes (and entire homes held out in cross-validation), a low-variance regularized linear model generalizes to an unseen home better than the higher-variance tree ensembles or nearest-neighbour rule, which overfit the small sample; this is the bias–variance trade-off favouring the simpler model on scarce, smooth data. The practical lesson is twofold: using an imputer matters far more than which one (one versus none is the gap that dominates), and on data of this size and structure a simple ridge regressor is the most accurate and lowest-risk choice, while the tree ensembles remain robust defaults that would scale to the much larger, less linear populations a utility ultimately faces.

E. NETWORK VALIDATION

The closed-loop contribution is summarized in Figure 7 and Table 6. The uncoordinated aggregate loads the 400 kVA transformer to 109.6% (past the 105% short-duration limit, a thermal violation), while the worst minimum voltage sits at a healthy 0.963 p.u. Ideal coordination clears the overload: maximum transformer loading falls to 99.0% (no violation, a healthy $\approx$ 6-point margin below the 105% limit) and the worst minimum voltage stays healthy at 0.967 p.u. This is the core result: flattening the aggregate is not merely a statistical improvement, it restores feeder feasibility.

The cleared state is resilient to communication failure through $\varrho = 0.8$, where the transformer holds at 102.1% loading and does not violate the 105% limit. The deterministic crossing falls between $\varrho = 0.85$ (104.4%, cleared) and $\varrho = 0.88$ (105.2%, violated), a critical fraction of ≈ 0.87; beyond it the loading ramps (106.2%) back toward the uncoordinated 109.6% once every home is imputed at $\varrho = 1$. The story is therefore $109.6\% \rightarrow 99.0\% \rightarrow 109.6\%$: coordination clears the overload, the clearance survives heavy non-responsiveness, and then the protection unwinds as the controllable population vanishes. This pins the critical

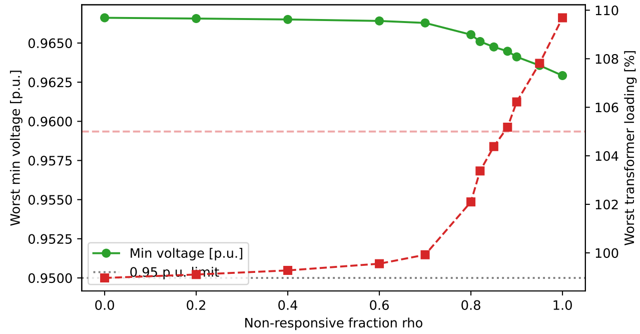

FIGURE 7. Worst minimum bus voltage (left axis) and worst maximum transformer loading (right axis) versus the non-responsive fraction ϱ, the headline feasibility figure. Loading stays cleared below the 105% limit through $\varrho = 0.85$, crosses it near $\varrho = 0.87$ (violated by $\varrho = 0.88$), and ramps to the uncoordinated 109.6% at $\varrho = 1$. The image's "rho" axis is the non-responsive fraction ϱ.

TABLE 6. Network feasibility on the LV feeder (worst of day), selected fractions, against the 105% short-duration limit. Loading is the 400 kVA transformer's worst-of-day loading.

Scenario	Peak (kW)	$V_{\min}$ (p.u.)	Loading (%)	Violation
Uncoordinated	401.6	0.963	109.6	yes
Coordinated ideal	363.8	0.967	99.0	no
ϱ=0.4	364.9	0.967	99.3	no
ϱ=0.6	365.9	0.966	99.6	no
ϱ=0.7	367.2	0.966	99.9	no
ϱ=0.8	375.0	0.966	102.1	no
ϱ=0.85	383.1	0.965	104.4	no
ϱ=0.88	385.9	0.964	105.2	yes
ϱ=0.9	389.6	0.964	106.2	yes
ϱ=1.0	401.9	0.963	109.7	yes

non-responsive fraction, beyond which imputation can no longer protect the transformer, to a narrow band around $\varrho = 0.87$.

a: Voltage stays healthy.

Across every scenario the worst minimum LV bus voltage stays in a comfortable 0.963–0.967 p.u. band and never approaches the 0.95 p.u. limit, so the feeder is voltage-comfortable throughout and the transformer thermal loading is the unambiguous binding constraint, a cleaner story than a benchmark that is stressed on voltage to begin with. The 105% short-duration loading limit gives an unambiguous binary verdict, and coordination is what moves the feeder across it from infeasible to feasible.

F. SOLVER VALIDATION

To rule out an artifact of the hand-rolled solver, the distributed ADMM was checked against a centralized CVXPY solve of the same convex program. The two agree on the aggregate peak, the quantity the feasibility verdicts rest on, to within 0.05%; the objective is reproduced exactly for the separable (energy-only) case and to within about 0.2% for the temperature-coupled case. Figure 8 shows the sharing-ADMM objective on the ideal coordination problem driving down to the CVXPY optimum within a few tens of iterations

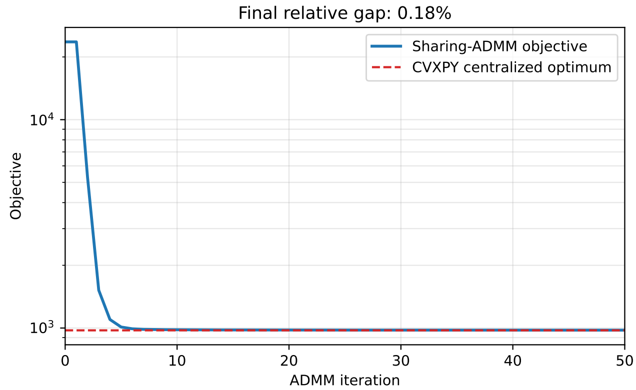

FIGURE 8. Sharing-ADMM objective (log scale) versus iteration on the ideal coordination problem, converging to the CVXPY centralized optimum (dashed); final relative gap 0.18%.

and settling at a 0.18% relative gap (the small residual is the proximal temperature-coupled x-update, and it shrinks with the agent count). The reported verdicts therefore reflect the optimization, not the implementation.

G. FORECAST UNCERTAINTY QUANTIFICATION

The network verdict above rests on a single realization per non-responsive fraction. To test whether that point estimate understates risk, we run the Monte-Carlo analysis of Section IV (200 draws per fraction over which agents fall silent and Gaussian forecast residuals of standard deviation 0.304 kW, the imputer's cross-validated RMSE), mapping each draw's daily peak to the worst-of-day transformer loading.

Figure 9 and Table 7 report the result (mean with [P5, P95] band). The probability of a thermal violation of the 105% limit stays at 0 through $\varrho = 0.8$, ticks to 0.01 at $\varrho = 0.82$, rises to 0.21 at $\varrho = 0.85$ and 0.81 at $\varrho = 0.88$, and is 1.00 for $\varrho \geq 0.90$. The single-realization analysis reported $\varrho = 0.85$ as compliant (104.4% loading), but once the uncertainty in which agents fail and in the forecast is accounted for, that operating point already breaches in 21% of draws, rising to 81% by $\varrho = 0.88$: the deterministic estimate understated how abruptly the risk rises. The [P5, P95] band is tight on the plateau (forecast errors average out across a large responsive population) and widens at the knee ($[103.6, 105.5]\%$ at $\varrho = 0.85$, $[104.7, 106.5]\%$ at $\varrho = 0.88$) before riding the ramp upward. This reframes the operator takeaway: the critical transition is not a single deterministic fraction but a probabilistic risk threshold centered near $\varrho = 0.87$, where the breach probability swings from near zero to near certainty across the sweep.

H. DISCUSSION

Three observations tie the results together. First, comfort, peak, and feasibility are governed by the same flexibility-budget mechanism: the responsive homes' collective $\pm 50\%$ deferral budget is what simultaneously bounds the indoor-temperature excursion, holds the aggregate peak near its optimum, and keeps the transformer cleared, so all three stay benign on the plateau and degrade together as that budget is exhausted past the knee near $\varrho = 0.87$. Second, the network result is not a restatement of the aggregate result: a 9.4% peak cut is a statistical improvement, but clearing a 109.6% transformer overload to 99.0% under the 105% short-duration limit is an operational state change from infeasible to feasible, and the two need not coincide on a different feeder or load allocation. Third, the unwinding past the knee is graceful rather than catastrophic (the loading ramps to 109.6% and the power flow always converges, never collapsing), which indicates that the imputer keeps the system in a recoverable regime even when most agents are silent. Together these support the central claim that coordination quality should be measured by feeder feasibility, and that imputation extends the range of communication failure over which that feasibility holds, here up to a critical non-responsive fraction near $\varrho = 0.87$.

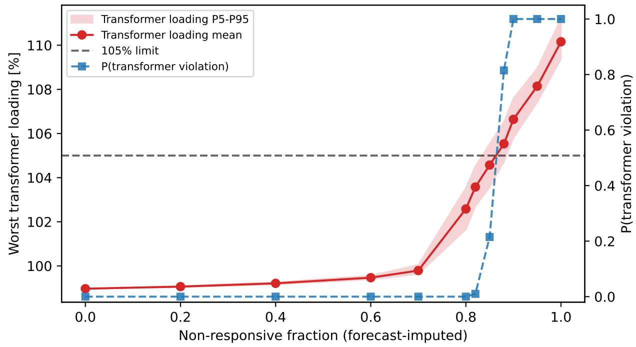

FIGURE 9. Monte-Carlo forecast-uncertainty sweep (200 draws per fraction). Worst maximum transformer loading mean with its P5–P95 band versus the non-responsive fraction ϱ (left axis, with the 105% short-duration limit), and the probability of a transformer violation (dashed line, right axis). The image's "rho" axis is the non-responsive fraction ϱ.

TABLE 7. Monte-Carlo transformer loading and violation probability by fraction (200 draws each; mean [P5, P95]).

ϱ	Loading (%) [P5, P95]	P(violation)
0.70	99.8 [99.6, 100.1]	0.00
0.80	102.6 [101.7, 103.5]	0.00
0.85	104.6 [103.6, 105.5]	0.21
0.88	105.5 [104.7, 106.5]	0.81
0.90	106.6 [105.8, 107.6]	1.00
1.00	110.2 [109.4, 111.2]	1.00

VII. CONCLUSION

We presented a network-validated pipeline that coordinates cold-climate electric heating with comfort-aware sharing-ADMM, reconstructs non-responsive agents with a machine-learning imputer, and, as the closing contribution, validates the coordinated aggregate on a synthetic LV residential feeder of 74 all-electric homes behind a 400 kVA distribution transformer with AC power flow. The closed loop turns abstract flattening metrics into a feeder verdict: ideal coordination cuts the aggregate peak by 9.4% and clears a 109.6% transformer thermal overload down to 99.0% under a 105% IEEE C57.91 short-duration limit, while LV voltage stays healthy (≈ 0.96 p.u., never approaching the 0.95 p.u. limit), so the transformer is the unambiguous binding constraint. Coordination also respects thermal comfort: propagating the schedules through each home's RC thermal model holds the worst modelled indoor-temperature excursion to $\approx 1.0\,^{\circ}\mathrm{C}$, against $2.1\,^{\circ}\mathrm{C}$ for a naive energy-only flatten. The cleared state is robust to communication failure through a non-responsive fraction of about 0.8, and a finely-resolved sweep shows the protection then unwinds as a plateau followed by a ramp: the worst transformer loading stays cleared below the 105% limit up to $\varrho \approx 0.85$, crosses it near $\varrho = 0.87$, and rises to the uncoordinated 109.6% as the last homes fall silent. A Monte-Carlo analysis over which agents fail and the forecast error reframes this critical fraction as a probabilistic risk threshold rather than a single deterministic point: the probability of a thermal violation stays near 0 through $\varrho = 0.8$, rises to 0.21 at $\varrho = 0.85$ and 0.81 at $\varrho = 0.88$, and reaches certainty by $\varrho = 0.90$, so the deterministic compliance at $\varrho = 0.85$ understated the true risk. Comparing three families of ML regressors for the imputer, we found that a regularized linear model (ridge) was the most accurate and lowest-risk on this dataset, because a low-variance linear model suits the small, smooth, near-linear relationship between heating and the weather/calendar/level features better than higher-variance tree ensembles or nearest neighbours. The decisive factor, however, was imputing at all versus not, the choice of method being secondary once any reasonable imputer is in the loop. The distributed ADMM was validated against a centralized CVXPY solve (aggregate peak agreeing to within 0.05%).

The present study isolates the coordination-and-feasibility loop on a controlled, fully reproducible testbed, and several natural extensions would broaden its reach. The results are demonstrated on a representative cold day and a single LV feeder, so the specific thresholds (the critical non-responsive fraction and the cleared loading margin) are best read as indicative for this configuration; a multi-feeder, multi-climate, and multi-day campaign is the natural way to map how they vary with topology and load mix. Thermal comfort is captured through a first-order RC penalty, a deliberately transparent physical model that higher-order or field-calibrated building models can refine. The communication-failure model, a stochastic non-responsive fraction applied at coordination time, cleanly isolates the effect of missing agents and can be enriched with a protocol-level treatment of delays, dropped packets, and retransmissions. The synthetic inputs are what make the pipeline reproducible end to end, and calibrating them against field measurements is a clear next step. We also plan to study the spatial placement of agents across the LV feeders, which governs how much each contributes to the binding constraint, and to couple the coordinator to a flexibility market so the cleared headroom can be priced and settled.

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